HK IMO Training 2018-2019 Problem Set 4

Suggested Solutions

Problem 17. The director has found out that six conspiracies have been set up in his department, each of them involving exactly three persons. Prove that the director can split the department into two laboratories so that none of the conspiring groups is entirely in the same laboratory.

Solution. Let the department consist of n persons. Clearly $n > 4$ (because $\binom{4}{3} < 6$). If $n = 5$, take three persons who do not make a conspiracy and put them in one laboratory, the other two in another. If $n = 6$, note that $\binom{6}{3} = 20$, so we can find a three-person set such that neither it nor its complement is a conspiracy; this set will form one laboratory. If $n \geqslant 7$, use induction. We have $\binom{n}{2} \geqslant \binom{7}{2} = 21 > 6 \cdot 3$, so there are two persons A and B who are not together in any conspiracy. Replace A and B by a new person AB and use the inductive hypothesis; then replace AB by initial persons A and B. □

Problem 18. A photographer took some pictures at a party with 10 people. Each of the 45 possible pairs of people appears together on exactly one photo, and each photo shows/depicts two or three people. What is the smallest possible number of photos taken?

Answer. 19.

Solution. Let x be the number of triplet photos (depicting three people, that is, three pairs) and let y be the number of pair photos (depicting two people, that is, one pair). Then $3x + y = 45$.

Each person appears with nine other people, and since 9 is odd, each person appears on at least one pair photo. Thus $y \geqslant 5$, so that $x \leqslant 13$. The total number of photos is $x + y = 45 - 2x \geqslant 45 - 2 \cdot 13 = 19$.

On the other hand, 19 photos will suffice. We number the persons $0, 1, \ldots, 9$, and will proceed to specify 13 triplet photos. We start with making triplets without common pairs of the persons 1 to 8:

$$123, 345, 567, 781$$

Think of the persons 1 to 8 as arranged in order around a circle. Then the persons in each triplet above are separated by at most one person. Next we make triplets containing 0, avoiding previously mentioned pairs by combining 0 with two people among the persons 1 to 8 separated by two persons:

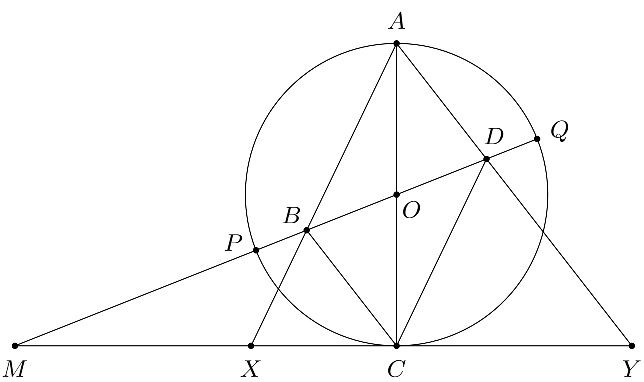
$$014, 085, 027, 036$$

Then we make triplets containing 9, again avoiding previously mentioned pairs by combining 9 with the other four possibilities of two people among 1 to 8 being separated by two persons:

$$916, 925, 938, 947$$

Finally, we make our last triplet, again by combining people from 1 to 8: 246. Here 2 and 4, and 4 and 6, are separated by one person, but those pairs were not accounted for in the first list, whereas 2 and 6 are separated by three persons, and have not been paired before. We now have 13 photos of 39 pairs. The remaining 6 pairs appear on 6 pair photos. □

Problem 19. Let $ABCD$ be a parallelogram. The circle with diameter AC intersects the line BD at points P and Q. The perpendicular to the line AC passing through the point C intersects the lines AB and AD at points X and Y, respectively. Prove that the points P, Q, X and Y lie on the same circle.



Solution. If the lines BD and XY are parallel the statement is trivial. Let M be the intersection point of BD and XY.

By Intercept Theorem

$$\frac{MB}{MD} = \frac{MC}{MY} \quad \text{and} \quad \frac{MB}{MD} = \frac{MX}{MC},$$

hence $MC^2 = MX \cdot MY$. Since MC is tangent to the circle, $MC^2 = MP \cdot MQ$. Therefore we have $MX \cdot MY = MP \cdot MQ$ and the quadrilateral $PQYX$ is inscribed. $\square$

Problem 20. Determine all positive integers $n \geqslant 3$ such that the inequality

$$a_1a_2 + a_2a_3 + \ldots + a_{n-1}a_n + a_na_1 \leqslant 0$$

holds for all real numbers $a_1, a_2, \ldots, a_n$ which satisfy $a_1 + a_2 + \ldots + a_n = 0$.

Answer. $n = 3$ and $n = 4$.
Solution. For $n = 3$ we have

$$a_1a_2 + a_2a_3 + a_3a_1 = \frac{(a_1 + a_2 + a_3)^2 - (a_1^2 + a_2^2 + a_3^2)}{2} \leqslant \frac{(a_1 + a_2 + a_3)^2}{2} = 0.$$

For $n = 4$, we have

$$a_1a_2 + a_2a_3 + a_3a_4 + a_4a_1 = (a_1 + a_3)(a_2 + a_4) \leqslant \frac{(a_1 + a_2 + a_3 + a_4)^2}{2} = 0.$$

For $n \geqslant 5$ take $a_1 = -1$, $a_2 = -2$, $a_3 = a_4 = \cdots = a_{n-2} = 0$, $a_{n-1} = 2$, $a_n = 1$. This gives

$$a_1a_2 + a_2a_3 + \ldots + a_{n-1}a_n + a_na_1 = 2 + 2 - 1 = 3 > 0.$$